Yumeng Xiu

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RESEARCG INTERESTS

- Robotics, Trajectory Prediction
- Learning to Control, Trustworthy AI Autonomy
- Optimization

EDUCATION

Carnegie Mellon University Pittsburgh, PA
Master of Science in Mechanical Engineering (Robotics Track) – Research Program 08/2020-05/2023(Expected)
GPA:3.93/4.0
Relevant Courses: *Numerical Methods in Engineering, Computer Vision, Machine Learning, Deep Learning, Optimization, Linear Control, Robot Localization and Mapping*

Beijing Institute of Technology Beijing, China
Bachelor of Science in Mechanical Engineering 08/2016-07/2020
Overall GPA: 3.5/4.0; Major GPA:3.86/4.0

RWTH Aachen University Aachen, Germany
Summer School, Major in Automation and simulation 07/2017
Relevant Courses: *Numerical Differentiation, Modeling of ODE, Discretization Methods, Nonlinear Equations*

PUBLICATIONS

1. Zhefan Xu, **Yumeng Xiu**, Xiaoyang Zhan, Baihan Chen, and Kenji Shimada. Vision-aided UAV Navigation and Dynamic Obstacle Avoidance using Gradient-based B-spline Trajectory Optimization. arXiv preprint arXiv:2209.07003, 2022.
2. Zhefan Xu*, Xiaoyang Zhan*, Baihan Chen, **Yumeng Xiu**, Chenhao Yang, and Kenji Shimada. A real-time dynamic obstacle tracking and mapping system for UAV navigation and collision avoidance with an RGB-D camera. arXiv preprint arXiv:2209.08258

RESEARCH EXPERIENCE

Large-Scale power system voltage control with Neural ISS Certificates 06/2022-Present
Advisor: Guannan Qu, Carnegie Mellon Univeristy

- Proposed ISS Lyapunov based neural compositional certificate framework for stabilizing large scale power systems, by treating a large system as an interconnection of smaller subsystems. Developed methods to find each subsystem a decentralized controller and an Input-to-State Stability (ISS) Lyapunov function. Each ISS Lyapunov function of subsystem could be collected to prove the global stability of power system.
- Developed Centralized Neural controller and Lyapunov functions to verify the global stablity in large-scale power systems. Developed decentralized neural controllers and Neural ISS Lyapunov functions that could be used across different subsystems.
- Utilized Pandapower tool for modelling and simulation of multiple power system cases. The Neural controllers and Lyapunov functions show great performance in voltage convergence and stability of power systems.

A real-time dynamic obstacle tracking and mapping system for UAV navigation and collision avoidance with an RGB-D camera 06/2022-09/2022
Advisor: Kenji Shimada, Computational Engineering and Robotics Lab, Carnegie Mellon Univeristy

- Developed a real-time dynamic obstacle tracking and mapping system for quadcopter obstacle avoidance using an RGB-D camera.
- Developed a region proposal detector that utilized depth-image to obtain obstacle region proposals for static environment representation. Applied Kalman filter to predict velocity of obstacles to separate static and dynamic ones. Developed a continuity filter to track and identify each dynamic obstacle.
- Developed a trajectory prediction algorithm for each dynamic object. Applied Markov chain rule to generate multiple future paths of each dynamic obstacle with Gaussian distribution, considering the collision between future trajectories and environment, the trajectory with highest probability of collison-free will be sampled.

Vision-aided UAV Navigation and Dynamic Obstacle Avoidance using Gradient-based B-spline Trajectory Optimization 05/2022-08/2022

Advisor: Kenji Shimada, Computational Engineering and Robotics Lab, Carnegie Mellon University

- Proposed a gradient-based B-spline path planner utilizing robot's onboard vision. Proposed a circle-based guide-point algorithm for static obstacles avoidance. Proposed a receding horizon distance field to prevent dynamic collisons based on vision-detected dynamic obstacles' future predictions. Applied the iterative re-guide strategy to generate the collision-free trajectory.
- Designed the CAD model of a customized drone. Established a ROS simulation platform based on real-world data. The algorithm shows great performance in simulation platform for generating a collison-free and smooth trajectory.
- Set up a real Unmapped Aerial Vehicle (UAV) drone platform. Conducted multiple physical experiments to verify the proposed planner in different scenarios, the planner successfully generates a smooth and collison-free trajectory in real time.

Dynamic Obstacle Detection and bounding box estimation 02/2021-05/2021

Advisor: Kenji Shimada, Computational Engineering and Robotics Lab, Carnegie Mellon University

- Applied YOLOv5 object detector to detect generated the 2D and 3D bounding box of each pedestrian. Applied Kalman Filter to track and identify each pedestrian.
- Developed a construction vehicle detector based on YOLOv5, generated training data for construction vehicles through real images and ROS simulation data.

Design of a Three-dimensional Crossed hole position Accuracy Detection Instrument 01/2020 - 07/2020

Advisor: Muzheng Xiao, Associate Professor at Department of Mechanical Engineering, Beijing Institute of Technology

- Designed a three-dimensional crosshole position degree visual inspection. Proposed a measurement method applicable to the crosshole position degree measurement accuracy.
- Applied NX UG modeling software to design and model the module structure. Utilized Abaqus software to complete finite element analysis of the column and displacement table.

SELECTED COURSE PROJECTS

Flexible Long Term Mortality prediction from Radiological Impressions 02/2022-05/2022

Instuctor: Bhiksha Raj, Language Technologies Institute, Carnegie Mellon Univeristy

- Utilized CNN for feature extractions of existing chest radiography image datasets, applied cox model to build a high quality survival analysis model.
- Applied Auton survival packages to perform experiments based on MNIST datasets and PLCO datasets. Utilized Discrete Cox Proportional Hazards model (DCPH) for Hazard survival analysis. The DCPH gaves good performance in mortality analysis.

SAGA: A Fast Incremental Gradient Method with Support for Non-Strongly Convex Composite Objectives 03/2022-05/2022

Instuctor: Guannan Qu, Department of Electrical and Computer Engineering, Carnegie Mellon Univeristy

- Implemented SAGA algorithm in strongly convex cases compared to GD, SGD, SAG and SVRG. SAGA achieves better convergence rates and small variance than SGD, SAG and SVRG, and less computation cost than GD.
- Developed a CNN neural network. Applied SAGA algorithm in the CNN network. The results shows SAGA algorithm is applicable to non-strongly convex problems without modification, the convergence rate is 10 times faster than SGD.

TECHNICAL SKILLS

Programming Languages: Python (Numpy, Pandas, Scipy), C/C++, Matlab

Framework/Tools: Pytorch, OpenCV, ROS, RViz, AutoCAD, SolidWorks, Git, ABAQUS, UG, CloudCompare

HONORS & AWARDS

2020 Beijing Institute of Technology FastGear Third Prize Scholarship 03/2020

Third Prize, Century Cup Creative Competition, BIT 05/2018

Beijing Institute of Technology the Second Prize Scholarship 09/2018,09/2019

Member of Student Science Association, BIT 2016-2017